



Heat-resistant memory technology

During beach days, smartphone shutdowns often occur due to electronic components failing in extreme heat, causing data loss as electrons destabilise and escape. Addressing this,



Researchers led by Deep Jariwala and Roy Olsson have developed First-of-its-kind high-temperature resistant memory device that can reliably store data at temperatures as high as 600° Celsius (<https://perntoday.upenn.edu>)

University of Pennsylvania researchers have developed a memory technology enduring temperatures up to 600°C, more than double commercial drives' threshold. Their device maintained stability for over 60 hours, setting a durability benchmark. It utilises ferroelectric aluminium scandium nitride (AlScN), unlike silicon-based memories failing around 200°C.

AlScN operates on a metal-insulator-metal structure, ensuring efficient operation despite erratic particle movement at high temperatures. This innovation promises sophisticated computing in extreme conditions, potentially revolutionising fields like deep earth drilling and space exploration, with ongoing research aiming for further integration and applications in artificial intelligence.

Breakthrough paves the way for efficient, durable polymer-based solar panels

A team from the University of Hong Kong's Mechanical Engineering Department achieved a breakthrough in organic photovoltaics (OPV), promising to transform sustainable energy solutions. OPV, using printable and eco-friendly polymer semiconductors, holds great potential for renewable energy. However, the challenge has been achieving both high efficiency and long-term stability due to the polymers' soft nature. Their research focused on Y6, an electron-accepting molecule, polymerised to enhance OPV device efficiency and stability. By controlling Y6 aggregation, they improved charge generation efficiency and maintained polymer blend morphology crucial for sustained performance. This discovery paves the way for efficient and durable polymer-based solar panels in various applications.

3D IC solution enables seamless data transmission in devices

UMC's recent breakthrough in semiconductor technology introduces a 3D IC solution tailored to radio frequency silicon

on an insulator (RFSOI) technology. Implemented on their 55nm RFSOI platform, this solution reduces die size by over 45% while maintaining optimal radio frequency (RF) performance, crucial for meeting the escalating bandwidth demands of 5G networks. Developed in Hsinchu, Taiwan, it utilises wafer-to-wafer bonding technology to vertically stack dies, resolving RF interference issues and enhancing functionality. This advancement addresses the complexity of RF front-end modules in modern devices, optimising space utilisation and facilitating seamless data transmission across diverse applications, including mobile, IoT, and virtual reality devices.

Origami-inspired soft robot

Researchers from Princeton and North Carolina State University have combined traditional paper-folding methods with advanced materials science to create a new soft robot. This robot navigates complex mazes with enhanced flexibility



The robot's segments can flatten and extend into cylinders (Photo by Frank Wojciechowski)

and manoeuvrability, integrating its steering system within its structure for precise control without sacrificing softness. Developed in specialised

labs at Princeton and NC State, the robot employs modular, origami-patterned cylindrical segments equipped with electrothermal actuators and silver nanowire heaters for precise movement control. Each segment operates independently or connects for various applications like cargo transport or assembly tasks, offering potential for self-repair and adaptation. Despite speed limitations, ongoing research aims to enhance the robot's capabilities for diverse applications.

Microcapacitors that could power chips of the future

Lawrence Berkeley National Laboratory (Berkeley Lab) and UC Berkeley researchers have achieved record-breaking energy and power densities in microcapacitors, addressing challenges in on-chip energy storage for electronic devices. By engineering thin films of hafnium oxide and zirconium oxide via atomic layer deposition, they induced a negative capacitance effect, enhancing performance. These crystalline films, exhibiting ferroelectric or antiferroelectric properties, were finely tuned to create an unstable state, allowing easy polarisation with minimal electric fields. Thickened films with inserted aluminium oxide layers maintained instability, enabling increased energy storage capacity. Embedded into three-dimensional micro capacitor structures, these advanced microcapacitors promise to enhance energy storage in IoT sensors, edge computing systems, and AI processors.